Luning Wang

EDUCATION

Tsinghua University **Beijing, China**

Department of Electronic Engineering

B.E. in Electronic Information Science and Technology ***09/2020- 06/2024***

- Overall GPA: 3.76/4.0
- GRE: Verbal 155/ Quantity 170, Total 325, AW 4.0
- TOEFL: Reading 29/ Listening 26/ Speaking 22/ Writing 25/ Total 102
- Relevant Courses: Signals and Systems(4.0), Media and Cognition(4.0), Communications and Networks(4.0), Digital Image Processing(4.0), Fundamental of digital logic and processor(3.6), Database(3.6)

University of Michigan **Ann Arbor, US**

Department of Electrical and Computer Engineering

M.S in Electrical and Computer Engineering ***08/2024- 05/2026***

RESEARCH INTERESTS

- Efficient Deep Learning
- Natural Language Processing (Large Language Models)
- Multimodal Learning

PUBLICATIONS

- [ENLSP NeurIPS Workshop'24] **"CSKV: Training-Efficient Channel Shrinking for KV Cache in Long-Context Scenarios"**. **Luning Wang**, Shiyao Li, Xuefei Ning, Zhihang Yuan, Shengen Yan, Guohao Dai, Yu Wang.
- [ICML'24] **"Evaluating Quantized Large Language Models"**. Shiyao Li, Xuefei Ning, **Luning Wang**, Tengxuan Liu, Xiangsheng Shi, Shengen Yan, Guohao Dai, Huazhong Yang, Yu Wang.
- [Arxiv'24] **"A Survey on Efficient Inference for Large Language Models".** Zixuan Zhou, Xuefei Ning, Ke Hong, Tianyu Fu, Jiaming Xu, Shiyao Li, Yuming Lou, **Luning Wang**, Zhihang Yuan, Xiuhong Li, Shengen Yan, Guohao Dai, Xiao-Ping Zhang, Yuhan Dong, Yu Wang.
- [ENLSP NeurIPS Workshop'23] **"LLM-MQ: Mixed-precision Quantization for Efficient LLM Deployment"**. Shiyao Li, Xuefei Ning, Ke Hong, Tengxuan Liu, **Luning Wang**, Xiuhong Li, Kai Zhong, Guohao Dai, Huazhong Yang, Yu Wang.

RESEARCH EXPERIENCES

NICS Lab, Energy Efficient Computing Group (Tsinghua University) **Beijing, China**

Nanoscale Integrated Circuits and System Lab, Energy Efficient Computing Group (NICS-EFC) is leaded by Professor Yu Wang, in Electronic Engineering Department, Tsinghua University. The group is committed to the research of energy-efficient circuits and systems design methodology towards the Artificial Intelligence (AI) scenario: Multi-agent Reinforcement Learning Algorithm, Efficient and Robust DL system, Domain Specific Acceleration, and Multi-agent system.

Undergraduate Research Assistant, Advisor: Prof. Yu Wang, Dr.Xuefei Ning ***12/2023- 02/2024***

- **Project: Evaluation of Quantized Large Language Models**
  - Responsible for evaluating the effect of quantization on dialogue abilities and trustworthiness of LLMs, using mt-bench and adv-glue.
  - Responsible for the writing of the parts concerning dialogue abilities and trustworthiness in our paper, including the main text and case studies in the appendix.
  - Contributed to the rebuttal process by conducting experiments on the effect of quantization on tasks including alpaca_eval, vicuna-bench and wizardlm-bench. Our paper was accepted by ICML 2024.

Undergraduate Research Assistant, Advisor: Prof. Yu Wang, Dr.Xuefei Ning ***03/2023- 09/2023***

- **Project: Low-Bit Quantization with Mixed Precision for Large Language Models**
  - Conducted sensitivity tests on leading LLMs (e.g., OPT, LLaMa), gathering per-block and per-layer sensitivity data and generating informative charts to guide subsequent mixed-bit quantization strategies.
  - Contributed to the experimental evaluation of our grouping and reordering quantization strategy. Evaluated quantization losses under various reorder strategies and assessed losses for automatic and heuristic bit-width allocation, finally achieving an average bit-width of 2.8 bits.
  - Contributed to the writing of section "Introduction" and "Method" in our research paper, which was accepted by the Efficient Natural Language and Speech Processing Workshop with NeurIPS 2023.

INTERNSHIP EXPERIENCES

ByteDance **Beijing, China**

Established in 2012, ByteDance entered the scene with Toutiao in August 2012 and followed up with Douyin in September 2016. In 2017, it made a significant global impact by launching TikTok and later merging with Musical.ly. Today, ByteDance's TikTok is a dominant global platform for short-form mobile videos, contributing to its remarkable success in the mobile internet market.

Algorithm Intern, Advisor: Dr.Ning Wang ***09/2023- 01/2024***

- **Project: The Development of an Appeal Chatbot for TikTok Moderation System**
  - Developed the retrieval-augmented generation (RAG) component for the chatbot, employing technologies such as SBert Encoder and FAISS. This encoder-retriever segment attained a top-1 recall rate exceeding 50% on the OpenBookQA dataset.
  - Employed our RAG pipeline to enhance the generation of Large Language Models including baichuan, mistral, and gpt-3.5-turbo for QA tasks, and achieved an improvement of 10% ~ 20% in accuracy on the OpenBookQA dataset.
  - Contributed to the development of the explanation generation model for reasoning. Implemented tuning strategies, including supervised fine-tuning and in-context learning with curated prompts, attaining an F1 score surpassing 70% in identifying violations within TikTok's moderation data.

Infinigence AI **Beijing, China**

Infinigence AI is a technology company focusing on providing complete solutions for the AI 2.0 era. Our vision is to "unleash infinite computing power and make AGI accessible", and we are committed to becoming the "M×N" middle layer in the process of implementing big models, and injecting new momentum into the development of AI industry in China and even around the world.

Algorithm Intern, Advisor: Dr.Xuefei Ning ***02/2024- 06/2024***

- **Project: Training-Efficient Channel Shrinking for KV Cache in Long-Context Scenarios**
  - Designed and implemented a window-based token reduction algorithm using accumulated attention scores.
  - Designed and implemented an SVD-based channel reduction algorithm together with quantization-aware training, which has achieved the SOTA result for the compression of KV Cache on certain long-context tasks.
  - Responsible as the first author of the paper, which has been accepted by ENLSP NeurIPS Workshop 2024..

SKILLS

- **Programming Languages**：Proficient in Python, Matlab. Have fundamental knowledge of C/C++, C#, Verilog, SQL, etc.
- **Software Tools**：Proficient in Linux, Git, PyTorch, Latex, MySQL, etc.

HONORS&AWARDS

- Comprehensive Excellence Scholarship of Tsinghua University (Top 30% in major, 8000CNY) ***2022-2023***
- Tsinghua Friends-Toyota Scholarship (5000 CNY) ***2022-2023***
- First Prize in the 5th 'Huiye Cup' Software Design Competition (5000 CNY) ***2021-2022***
- Volunteer Excellence Scholarship of Tsinghua University (2000 CNY) ***2021-2022***